(i.e) when [S] >>> K_m;

$K_m + [S] = [S]$

Therefore, $V_{max} = \frac{(k_3[E_T][S])}{([S])} = k_3\ ET$

Substitute this value of $k_3[ET]$ in equation (4)

$$V = \frac{(V_{max}[S])}{(K_m + [S])} \qquad ...(5)$$

This equation is known as Michaelis - Menten equation.

Enzymes may be denatured by excessive heat, extreme pH or various chemicals. Such extremities or chemicals cause enzymes to denature by the disruption and possible destruction of both the secondary and tertiary structures. Such changes are irreversible.

9.1.1. Significance of Michaelis - Menten Equation:

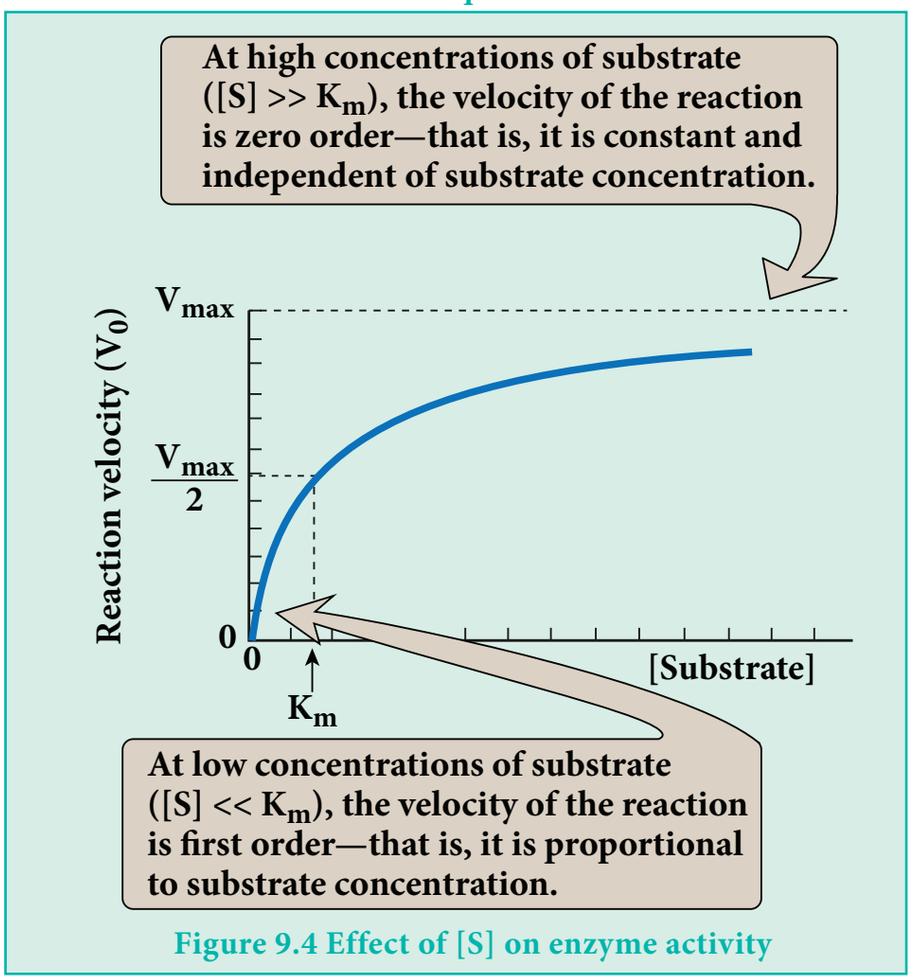

Figure 9.4 Effect of [S] on enzyme activity

Michaelis - Menten equation accounts for the three different phases of the substrate - concentration graph.

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